

①

Sol DES Problem:- DES (Data Encryption Standard) is a symmetric block cipher that operates on a 64-bit plain text using a 56-bit effective key. DES uses a feistel structure with 16 rounds

Step:-1 :- Initial permutation

64-bit divided into two 32-bit

$$IP(P.T) = L_0 || R_0$$

$L_0 \Rightarrow$ left 32 bits, $R_0 \Rightarrow$ right 32 bits

Step:-2 :- Expansion of R_0

32-bit $\rightarrow$ 48 bits

$$R_0 (32 \text{ bits}) \rightarrow E(R_0) (48 \text{ bits})$$

Step 3:- XOR with Round Key

$$X = E(R_0) \oplus K_1$$

Step 4:- S-box substitution

48-bit's $\rightarrow$ 32 bit's

Passed through a DES S-box

48 bits $\rightarrow$ 32 bits

Step 5 :- P-box permutation

32 bit are rearranged using the P-permutation

$$F(R_0, K_1) = P(s(E(R_0) \oplus K_1))$$

Step 6:- XOR with L_0

$$R_1 = L_0 \oplus F(R_0, K_1)$$

$$L_1 = R_0$$

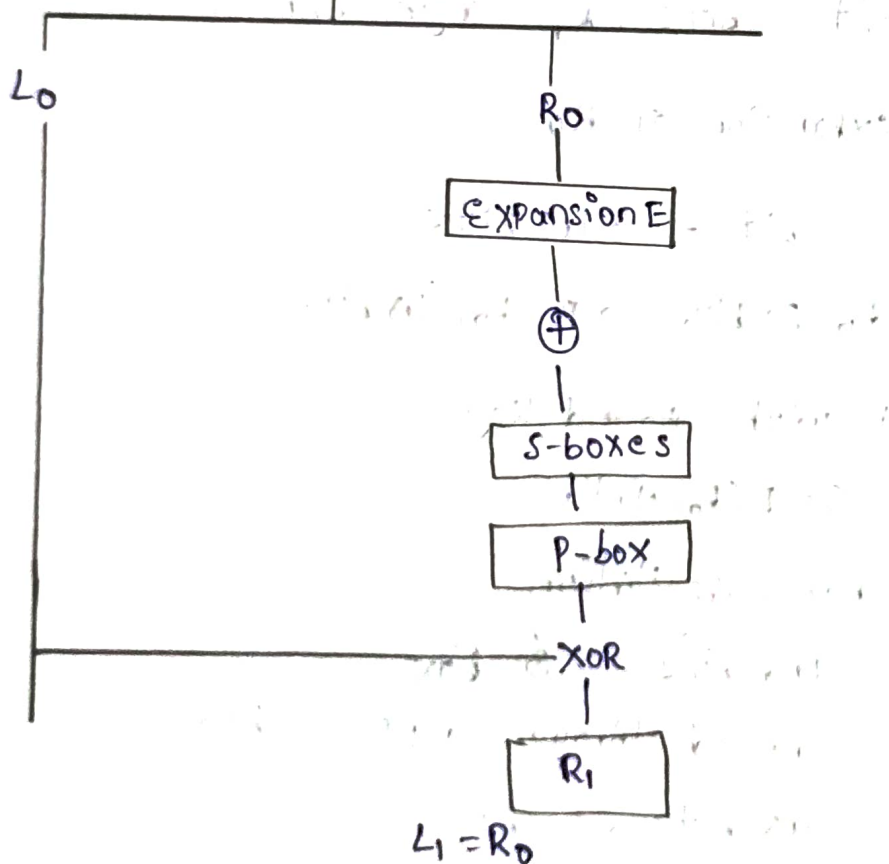
one DES round:

$$L_1 = R_0$$

$$R_1 = L_0 \oplus F(R_0, K_1)$$

plain-text

Initial permutation



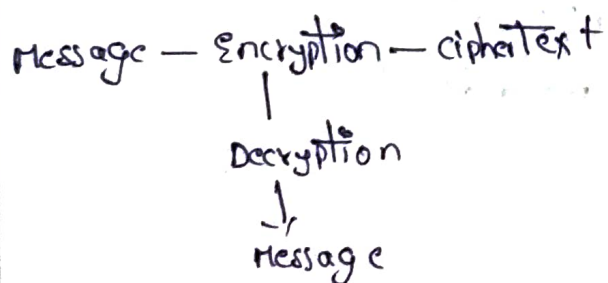
compare Symmetric and Asymmetric encryption using real-time applications.

Sol Encryption techniques are mainly classified into Symmetric encryption and asymmetric encryption.

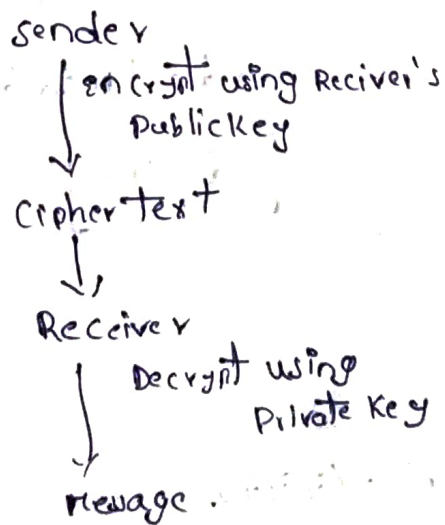
Feature	Symmetric Encryption	Asymmetric encryption
Keys used	one shared Key	Public Key and private Key
Encryption/Decryption	Same Key	Different Key
Speed	very fast	comparatively slow
Key distribution	Difficult	easier
Suitable for	Large amount of data	Key exchange & authentication
Example	AES, DES, Chacha20	RSA, ECC

Symmetric encryption

Secret Key



Asymmetric encryption



③

Sol

Given,

$$p = 7$$

$$q = 11$$

$$m = 9$$

$$n = p \times q$$

$$\Rightarrow 7 \times 11 = 77$$

$$\phi(n) = (p-1)(q-1)$$

$$= (7-1)(11-1) = 6 \times 10$$

$$\boxed{\phi(n) = 60}$$

Step 3: select public exponent

$$e = 7$$

$$\gcd(7, 60) = 1$$

$$\boxed{\therefore e = 7}$$

calculate private key:-

d

$$ed = 1 \pmod{60}$$

$$7d \equiv 1 \pmod{60}$$

$$\boxed{d = 43}$$

$$7 \times 43 = 301$$

$$301 \pmod{60} = 1$$

RSA Keys :-

Public Key :-

$$(e, n) = (7, 77)$$

private Key :-

$$(d, n) = (43, 77)$$

RSA encryption

$$C = M^e \bmod n \Rightarrow C = 9^7 \bmod 77$$

$$9^2 = 81 \equiv 4 \bmod 77$$

$$576 \bmod 77 = 37$$

$$\therefore C = 37$$

$\therefore$ encrypted ciphertext = 37

Q

Sol A CBC mode:- (Cipher block chaining)

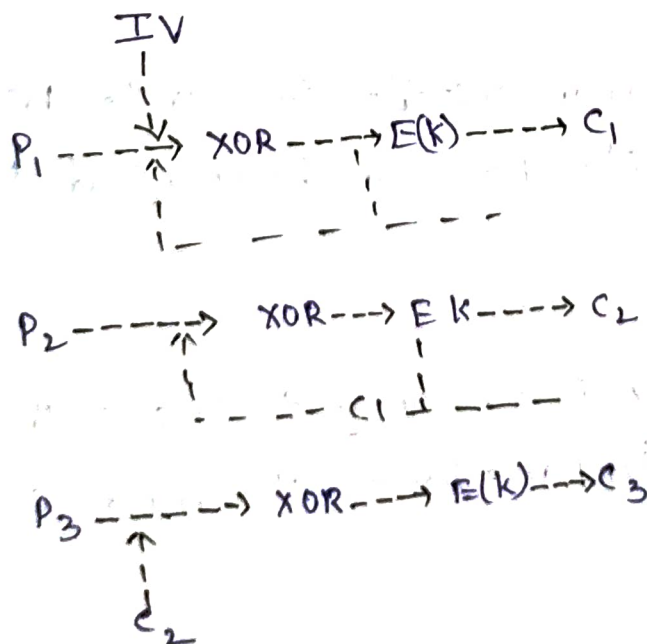
In CBC each plaintext block is XOR with the previous cipher block before encryption.

1st block $\rightarrow$ (IV) is used

Encryption

$$C_i = E_k(P_i \oplus C_{i-1}) \quad \because C_0 = IV$$

CBC :-



B. CTR Mode :- (Counter mode)

CTR converts a block cipher into a stream-like cipher.

$$C_i = P_i \oplus E_k(\text{Nonce || counter}_i)$$

CTR Encryption Diagram :

$$\text{Nonce + counter}_1 \rightarrow E(k) \rightarrow +$$

$$P_1 \xrightarrow{\hspace{2cm}} \text{XOR} \rightarrow C_1$$

$$\text{Nonce + counter}_2 \rightarrow E(k) \rightarrow +$$

$$P_2 \xrightarrow{\hspace{2cm}} \text{XOR} \rightarrow C_2$$

$$\text{Nonce + counter}_3 \rightarrow E(k) \rightarrow +$$

$$P_3 \xrightarrow{\hspace{2cm}} \text{XOR} \rightarrow C_3$$

- Does not require chaining between plaintext blocks.
- Blocks can be processed independently.
- Suitable for high-speed applications.

which is better for parallel processing

$\Rightarrow$ CTR Mode

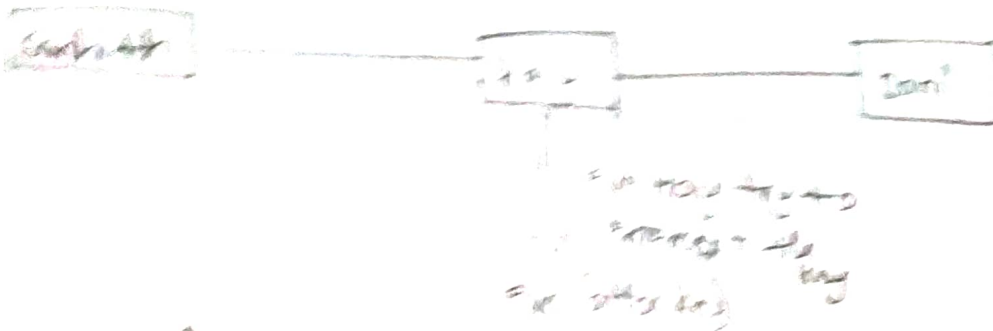
CTR is more suitable for parallel processing because each counter block can be encrypted independently

⑤ Quantum cryptography :

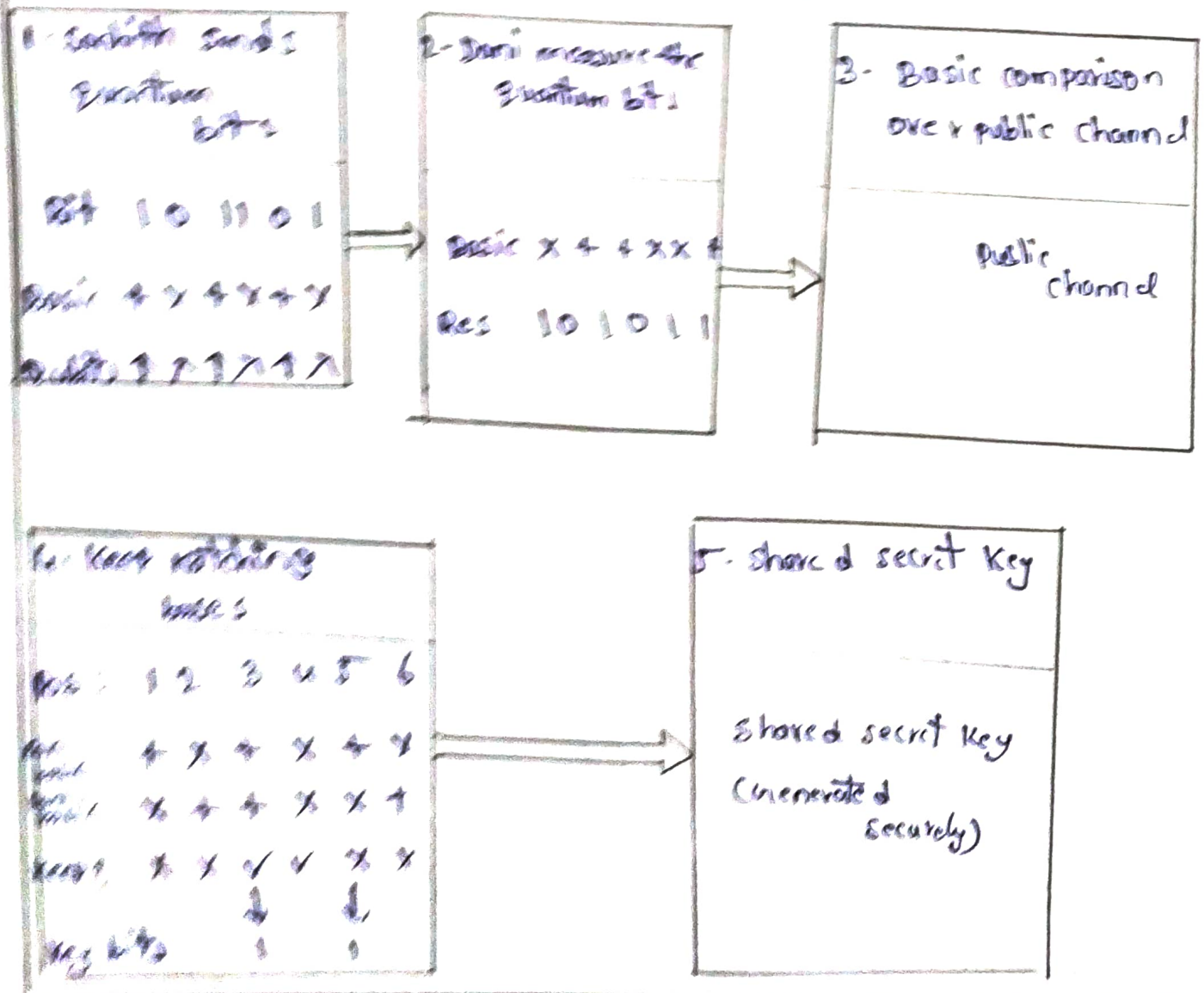
Quantum cryptography is a security technique that uses the principles of quantum mechanics to provide secure communication.

* The most important application of quantum is quantum key distribution (QKD).

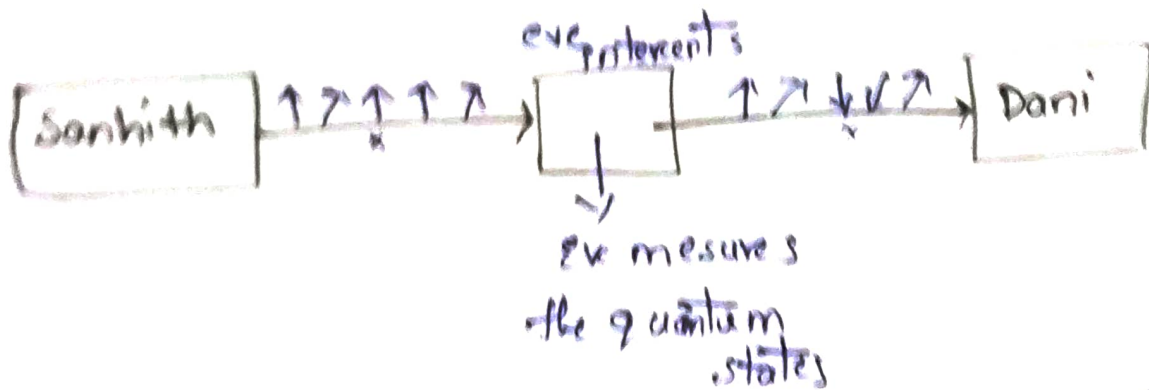
Quantum Key Distribution



How Quantum Key Distribution (QKD) works:-



if Eve tries to attack:



Because Eve measured the qubits, the quantum states get disturbed (changed randomly). This introduces error.